VIT
Vellore Institute of Technology
CHENNAI

Reg. Number:

Continuous Assessment Test (CAT) – I - JAN 2026

Programme	:	B.Tech	Semester	:	Winter Semester 2025-26
Course Code & Course Title	:	BCSE204L - Design and Analysis of Algorithms	Class Number	:	CH2025260501955 CH2025260501953 CH2025260501958
Faculty	:	Dr. Rajakumar Arul Dr. M. Dominic Savio Dr. K. Valarmathi	Slot	:	E2+TE2
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1	i.	Given an n-digit positive integer p of n distinct digits, where each digit $d_i \in \{0, 1, \ldots, 9\}$ and $k > 0$. It is assumed that p contains digits that occur only once. Design a brute force algorithm to generate all permutations p' of p. **Example:** Let p=153. The permutations of p are: 153, 135, 513, 531, 315, 351. [Pseudocode: **4 marks,** Time-complexity: **1 mark**]	5	CO 1	K3
	ii.	A perfect number is a number that is equal to the sum of its proper divisors excluding itself. Given an array of non-negative integers, write an algorithm to find the largest number in the array that is a perfect number. [Pseudocode: **4 marks,** Time-complexity: **1 mark**]	5	CO 1	K3
2		Furthest-in-Future Cache Replacement Algorithm: You are given a sequence of data requests and a cache of fixed size k. The cache is initially empty, and the requests are processed one after another in the given order. If a requested data item is already present in the cache, it is treated as a cache hit (H) and no replacement is required. If a requested data item is not present in the cache, it results in a cache miss (M). When a cache miss occurs and the cache still has free space, the requested item is simply inserted into the cache. However, when a cache miss occurs and the cache is already full, the cache manager must decide which existing item to remove. The replacement decision must follow the Furthest-in-Future strategy, in which the data item whose next request occurs farthest in the future, or which does not	10	CO1 & CO2	K4

		appear again in the remaining request sequence, is selected for removal. Write an algorithm that processes the entire request sequence and outputs the decision made for each request, clearly indicating whether the request results in a hit, an insertion, or the removal of a specific data item. Example ---------- Sequence of data requests: 2 3 5 8 2 3 4, cache size k = 3. Sequence of data requests: 2 3 5 8 2 3 4 Cache location 1: 2 2 2 2 2 2 4 Cache location 2: 3 3 3 3 3 3 Cache location 3: 5 8 8 8 8 Type of operation: M M M M H H M [Logic: **2 marks,** Pseudocode: **6 marks,** Time-complexity: **2 marks]**			
3		The distances between several pairs of points along a straight line are expressed in English units, namely feet and inches. Each distance is represented as an ordered pair (f, i), where f denotes the number of feet and i denotes the number of inches ($0 \leq i < 12$). Given a set of n such distances, design and apply a divide and conquer algorithm to determine both the smallest distance and the largest distance among them. The algorithm should divide the given set of distances into smaller subsets, recursively find the minimum and maximum distance in each subset, and then combine these results to obtain the overall smallest and largest distances. The final result should be reported clearly in terms of feet and inches. [Logic: **2 marks,** Illustration: **2 marks,** Pseudocode: **4 marks,** Time-complexity: **2 marks]**	10	CO1 & CO2	K4
4.		Given a chain of matrices with dimensions: $M_1: 4 \times 9,\ M_2: 9 \times 6,\ M_3: 6 \times 8,\ M_4: 8 \times 12$ Formulate a dynamic programming solution to compute the minimum number of scalar multiplications. Construct the DP cost table and determine the valid parenthesization for matrix M_1 to M_4. i) Construct the DP cost table (2 Marks) ii) Illustrate the process of calculating the minimum scalar multiplication step by step. (5 Marks) iii) Determine the valid parenthesization for matrix M_1 to M_4. (3 Marks)	10	CO2	K3
5		Let $X = x_1x_2 \ldots x_m$, $Y = y_1y_2 \ldots y_n$ be two strings over a finite alphabet Σ. A Longest Common Substring (LCSub) is defined as a longest substring (Z) that exists both in X and Y in continuous form (i.e. $Z = X[i..i+k-1] = Y[j..j+k-1]$ for some $1 \leq i \leq m,\ 1 \leq j \leq n$). Your task is to find the LCSub of $X = ABABC$ and $Y = BABCA$ using dynamic programming's approach. Write an algorithm to find the longest common substring that exists both in X and Y. [Logic: **2 marks,** Illustration: **2 marks,** Pseudocode: **4 marks,** Time-complexity: **2 marks]**	10	CO2	K6

**********All the best **********